

Metric switching: tapping analysis

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What this document is

Participants tapped along with a metronome. Partway through some trials, the underlying meter changed (a “switch”) – either **cued** (an external signal told them when) or **spontaneous** (no signal; if a switch happens, it's the participant's own doing). We recorded every tap's timing and loudness and want to know: how does tapping behaviour change around a switch, and can we estimate *when* a switch happened even when nobody told the participant to switch?

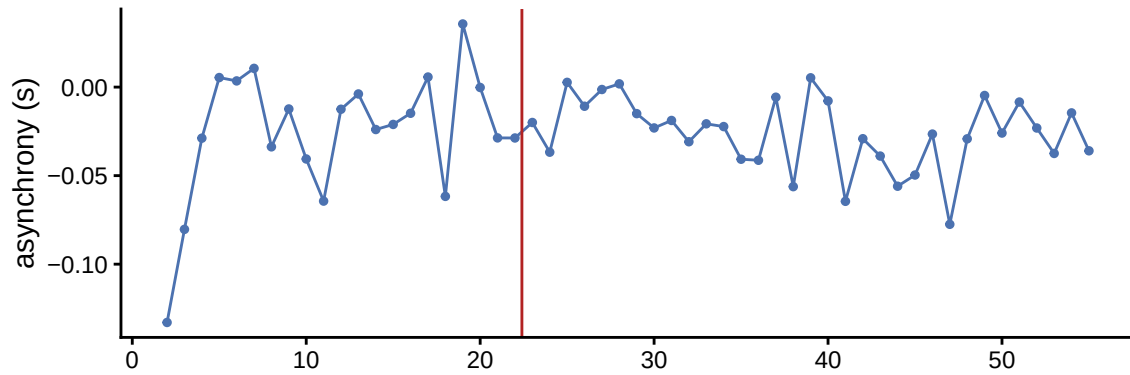
Currently based on one participant (**sub02**) with clean, low-noise recordings; a second participant's data was excluded (recording was too quiet/noisy to trust).

Part 1: One trial, up close

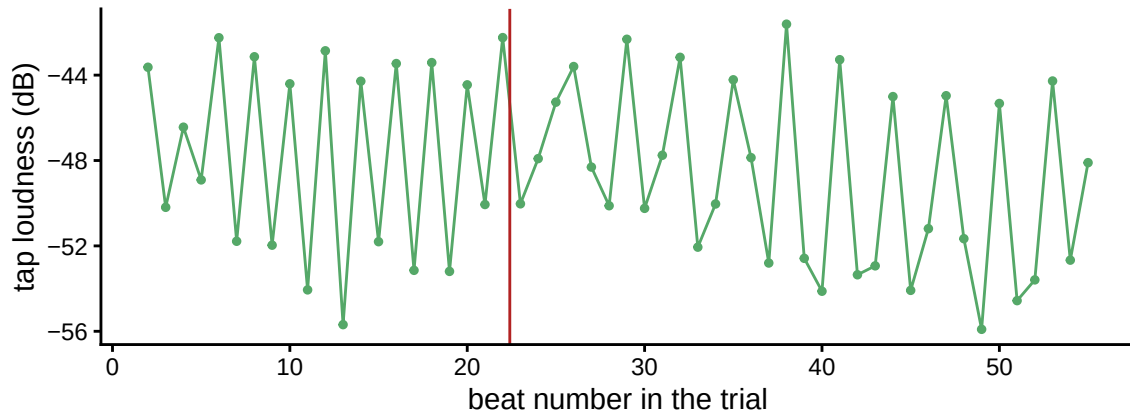
Before any pooling or modelling, here's a single cued trial in full: every tap's timing (top) and loudness (bottom), with the red line marking exactly when the participant was cued to switch meter.

Trial 15: timing

Red line = when the switch was cued



Trial 15: loudness

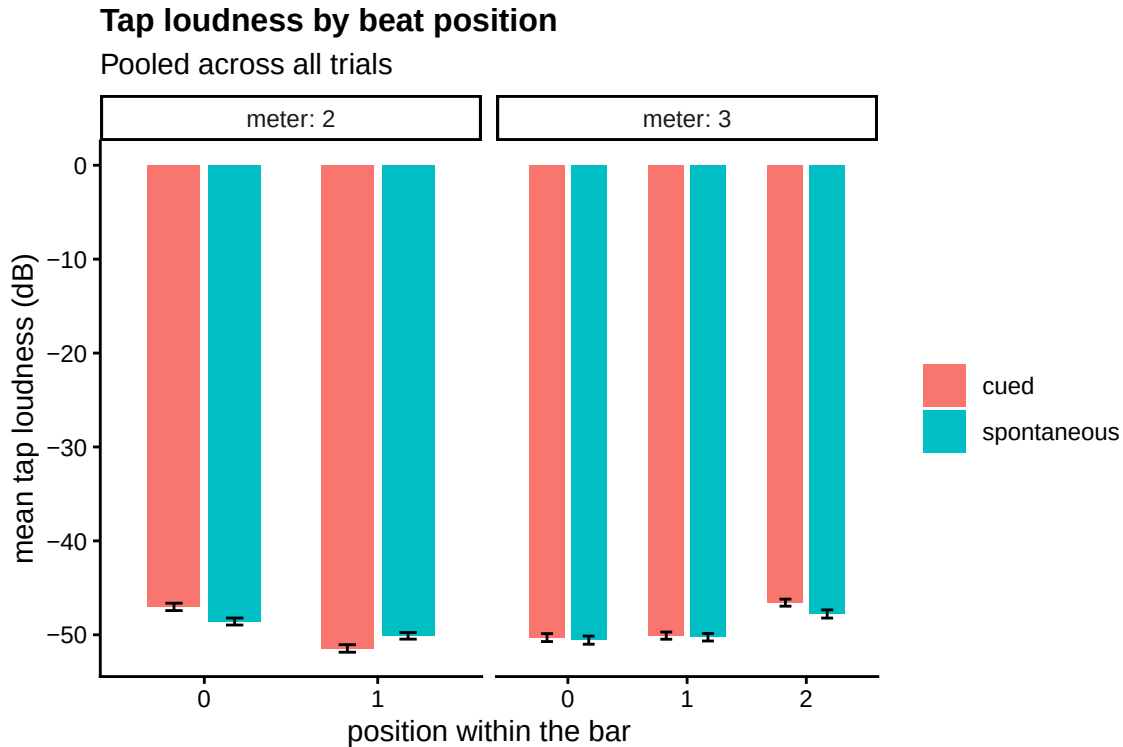


Look at the two panels around the red line. In the **timing** panel, it's hard to tell anything happened at all – the tap-to-click offset looks about as noisy before as after. In the **loudness** panel, though, there's a visible change in *pattern*: before the switch, every other beat is noticeably louder (a clear accent, alternating strong-weak); after it, that particular alternation is gone.

Is this trial typical?

Condition	Meter	N trials
cued	2	5
cued	3	5
spontaneous	2	5
spontaneous	3	5

20 trials total: 10 **cued** (an instruction tells the participant when to switch) and 10 **spontaneous** (no instruction), evenly split between the two meters tested (2 and 3 beats per bar). Tempo ranged 100-120 BPM across trials. Pooled across every trial, timing stays centred on zero with no obvious switch-related structure, confirming the single trial above wasn't a fluke – **timing alone doesn't show a reliable switch signal**.



Loudness, on the other hand, generalises: across trials and both meters there’s a consistent loudness contrast between beat positions (a louder “downbeat”). **This is why the model below is built entirely on loudness, not timing** – it’s the channel that actually carries the switch signal.

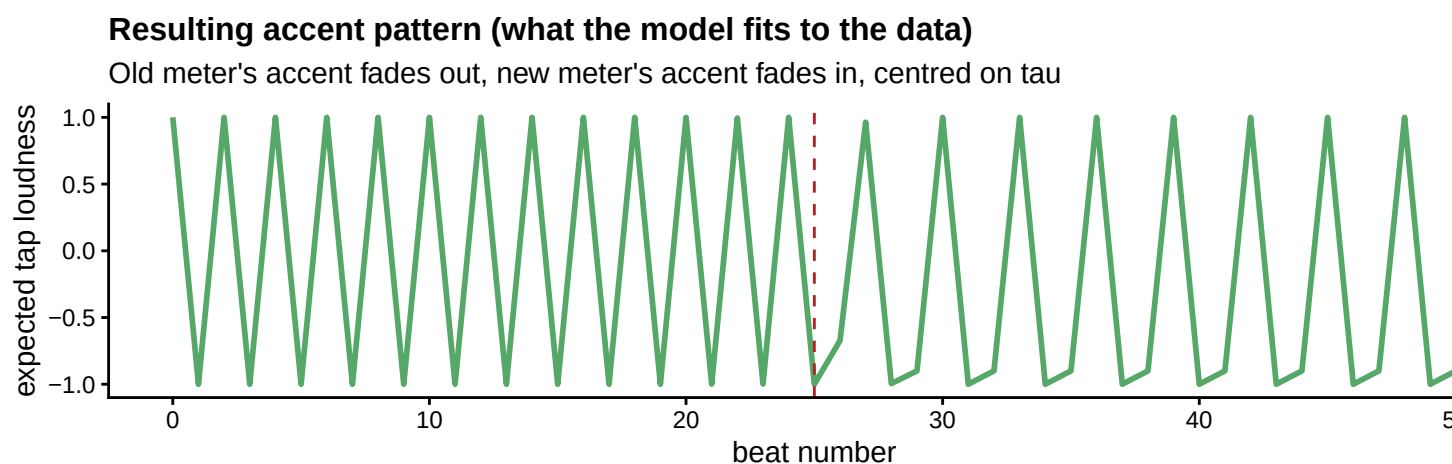
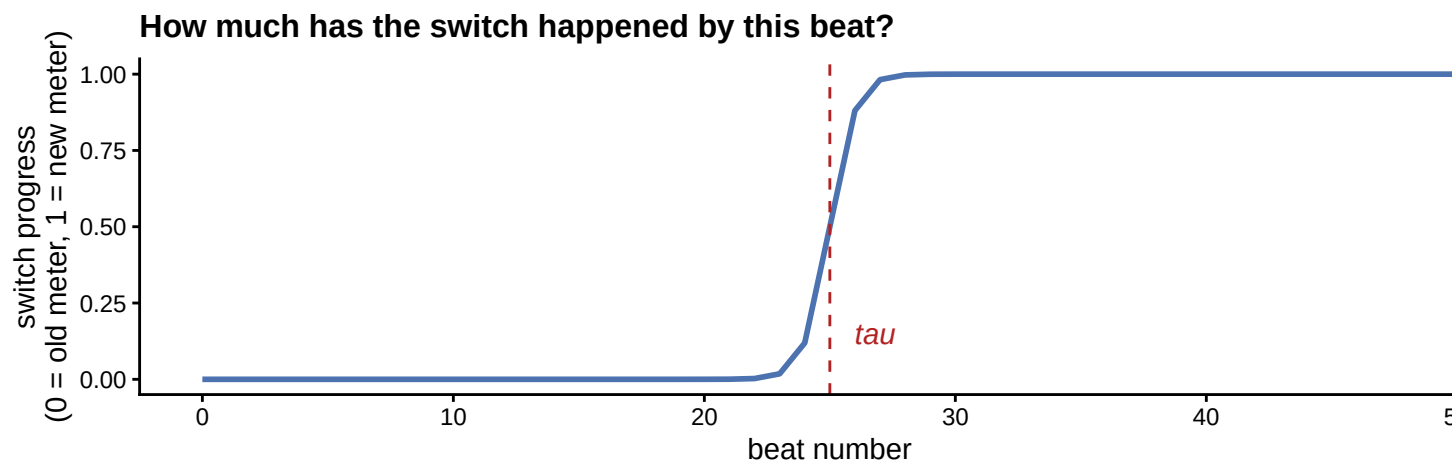
Part 2: What the model does

We fit a statistical model to each trial’s tap-loudness pattern. In plain terms, the model looks for **the point in the trial where the participant’s tapping pattern changes from one meter’s accent to the other’s**, and reports:

- **tau** – **when** the change happened, given as a beat number in the trial (or, for cued trials, we can compare it directly to when the instruction was given).
- **how confident** the model is in that estimate, from the *width* of tau’s credible interval – a trial with a clear, well-defined switch gives a narrow, tight estimate; a trial where the data don’t show a clear switch (or don’t show one clearly enough to pin down) gives a wide, uncertain estimate instead of a falsely precise one.

Concretely, each beat’s loudness (in dB) is converted to a log-amplitude number, and that’s the one value the model fits at every beat – expected to sit near one meter’s accent pattern before the switch and the other’s after it, blending smoothly in between. (Full formulas in `../docs/model_math.md`.)

The model allows the switch to be gradual, but doesn’t assume it has to be – checking this directly against the data (comparing a smooth-transition version of the model against a version where the switch is instantaneous) shows real switches are close to instantaneous: tapping style changes over roughly a beat, not gradually across many beats. Either way, the model only uses the loudness/accent information described in Part 1 – not raw timing – to find the change point.



(*Illustrative example, not a fit to real data.*) Top: the model’s “switch progress” curve – a smooth S-shape that goes from 0 (still in the old meter) to 1 (fully in the new meter), centred on τ . Bottom: this progress curve blends the old meter’s accent pattern into the new one’s, which is what the model actually compares against the real tap-loudness data. The model can fit either a narrow (quick) or wide (gradual) S-curve, whichever the data support – and as noted above, real trials consistently favour a narrow one, close to an instant switch.

We do **not** detail the model’s internal mechanics beyond this here; see `../docs/model_math.md` and `02_fit_model.ipynb` for the full specification.

Validation. Before trusting the model on the harder spontaneous case, we checked it against the cued trials, where we know when the *instruction* to switch was given. τ isn’t meant to match that instruction time exactly – it’s fit purely from tap loudness, so it measures when the participant’s *tapping itself* actually changed, which should come a little after the instruction (reaction time, plus a moment to settle into the new pattern). That’s exactly what we see below: across 9 of the 10 cued trials (one trial’s fit didn’t converge reliably and is excluded – flagged in the table), τ tracks the instruction time closely with a consistent positive lag, never negative.

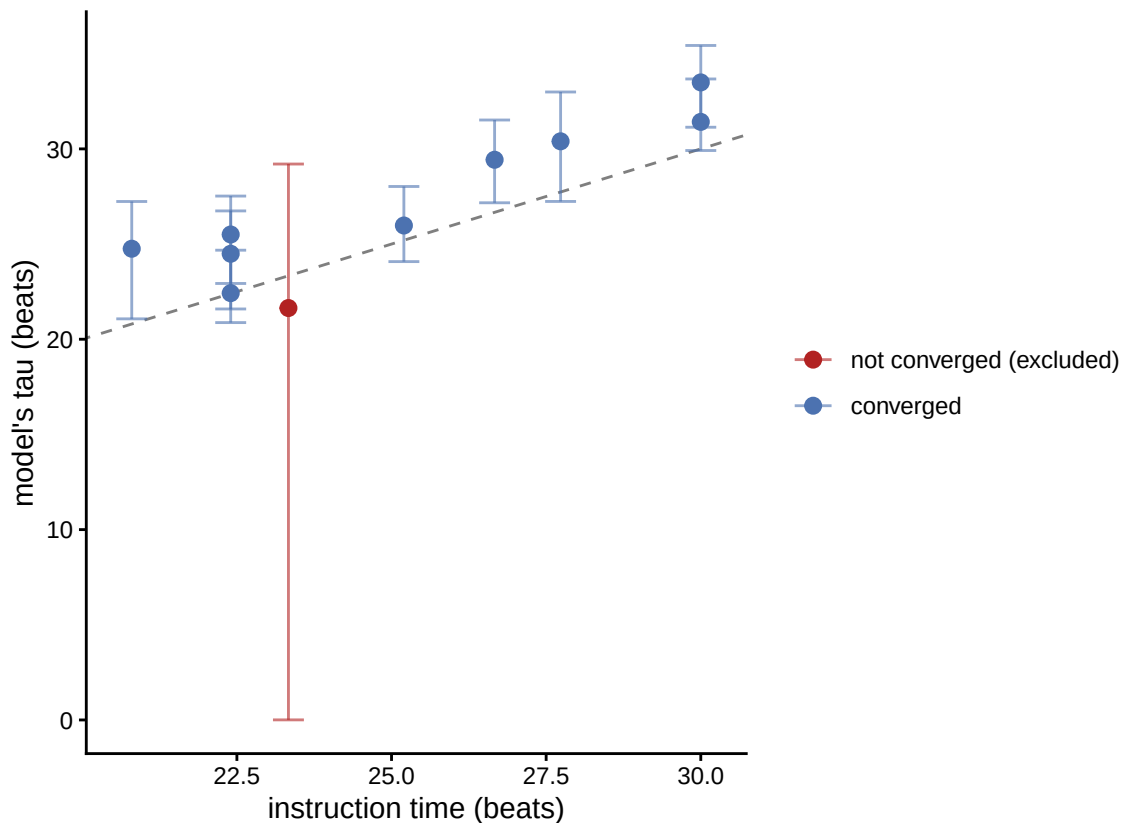
trial	meter	instruction time (beats)	model's tau (beats)	lag (tau - instruction)	converged?
3	3	25.2	26.0	0.8	yes
5	2	20.8	24.8	4.0	yes
6	3	30.0	31.4	1.4	yes
10	3	22.4	22.4	0.0	yes

trial	meter	instruction time (beats)	model's tau (beats)	lag (tau - instruction)	converged?
12	2	23.3	21.6	-1.7	NO – excluded below
14	3	27.7	30.4	2.7	yes
15	2	22.4	24.5	2.1	yes
17	3	22.4	25.5	3.1	yes
19	2	26.7	29.4	2.8	yes
20	2	30.0	33.5	3.5	yes

Model's tau vs. the actual cue time, per cued trial

Converged trials: correlation = 0.94, mean lag = 2.25 beats

Dashed line = perfect agreement (tau == instruction time)



Part 3: Spontaneous switches

With no cue, there's no obvious external marker to tell us when (or whether) a participant switched on their own. This is exactly where the model's **tau** estimate helps: for each trial, we can ask *when the model thinks the participant changed*, and *how sure it is* (from the width of **tau**'s credible interval).

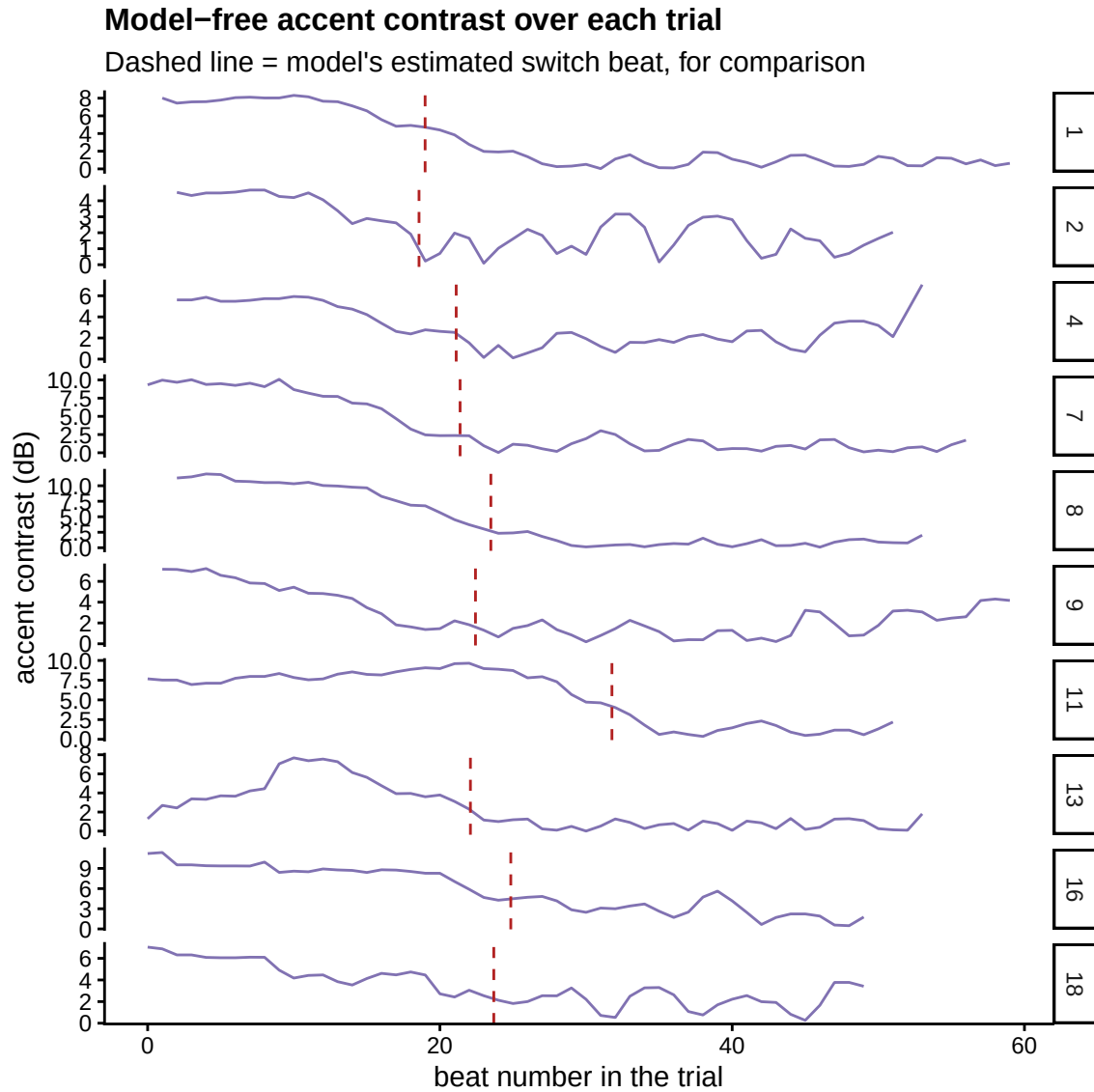
trial	estimated switch (beat)	94% CI width (beats)	model confidence
1	19.0	4.3	confident switch
2	18.6	5.2	confident switch
4	21.1	4.5	confident switch
7	21.4	4.1	confident switch

trial	estimated switch (beat)	94% CI width (beats)	model confidence
8	23.5	4.1	confident switch
9	22.4	8.7	confident switch
11	31.8	4.4	confident switch
13	22.1	12.2	confident switch
16	24.8	4.8	confident switch
18	23.7	5.6	confident switch

10 of the 10 spontaneous trials show a confident, well-identified switch; the rest are ambiguous (a wide `tau` credible interval means the data don't clearly pin down when – or whether – the participant's tapping pattern actually changed). We only use the confident trials below – mixing in the ambiguous ones would add noise from switch estimates the model itself is unsure about.

A model-free check: does the accent pattern itself show a switch?

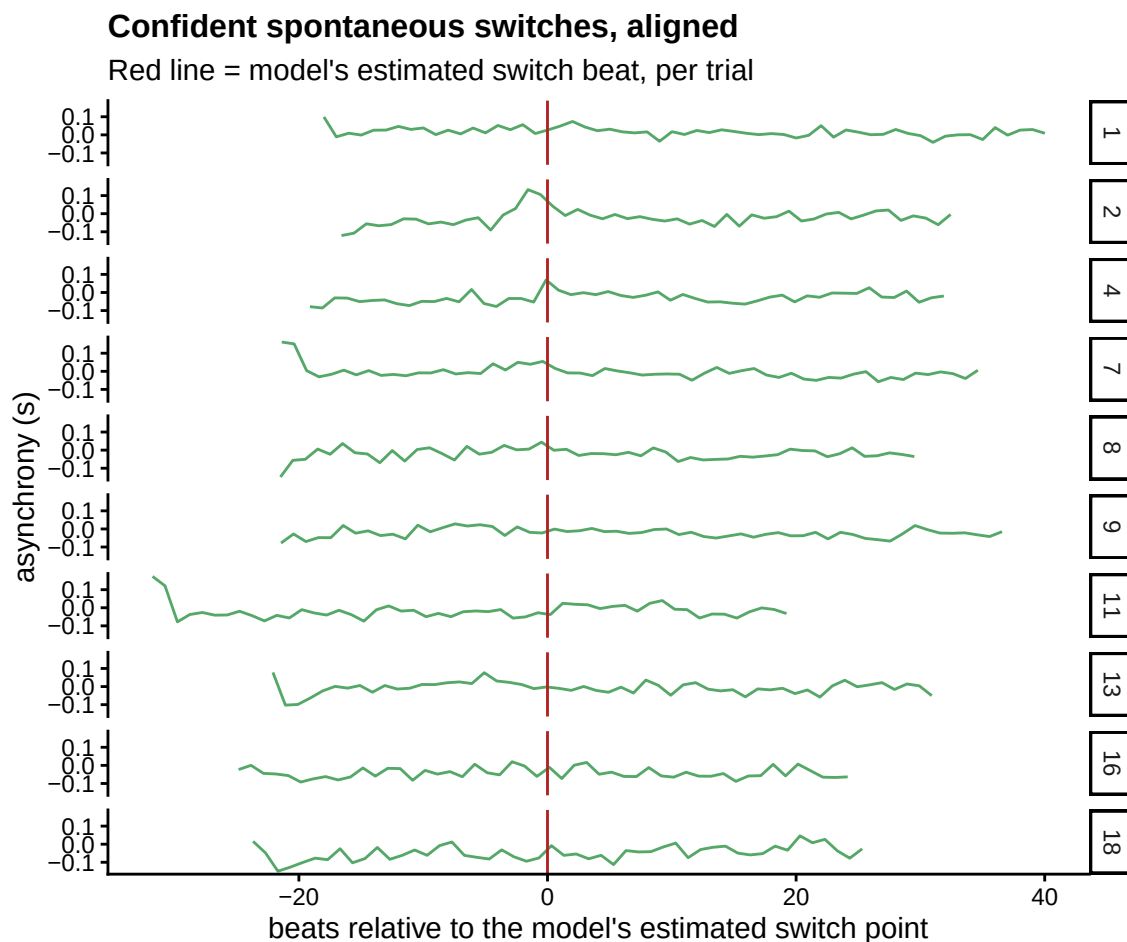
Before trusting the model's `tau` estimates, it's worth asking whether a switch is visible directly in the raw loudness data, with no fitting at all. For each trial, we slide an 8-beat window along the trial and compute one simple number: **how much loudness varies across the different positions in the bar, right now** (the gap between the loudest and quietest position, in dB, using only the taps in that window). If a participant has a clear accent pattern, this number is high; if it fades to a flat, unaccented tapping style, this number drops toward zero. A clear, sustained drop is a model-free signature of a switch.



Where the model reports a confident switch, the accent contrast typically starts high and shows a clear, sustained drop close to the model's estimated τ – a useful independent check that the model isn't just fitting noise. Note that individual taps still look “zig-zaggy” by eye even after a real collapse (beat-to-beat noise is large relative to the shrunk accent), so a raw per-tap plot alone can look similar before and after a switch – the rolling contrast statistic is what actually reveals the change, by averaging out that noise.

Where the model was unsure, there are two different reasons, visible directly in this plot: trials where the contrast **never gets very high to begin with** (e.g. trial 13, which starts under 2 dB) have no strong initial accent to switch away from, so there's nothing for the model to detect a collapse in; trials where the contrast drops only partway and not all the way to zero (e.g. trial 18) show a weaker, less complete change. Both are genuine features of the data, not a shortcoming of the fitting procedure – they're exactly why the model correctly reports low confidence there rather than forcing a precise answer.

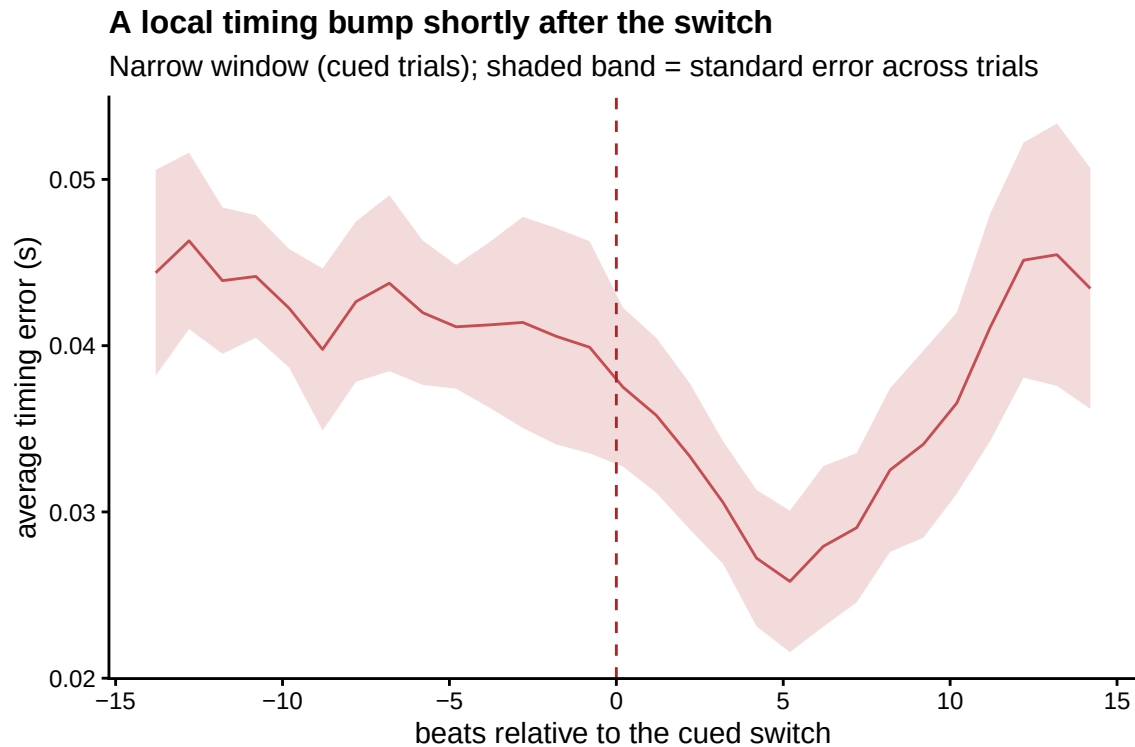
Aligning spontaneous trials on the model's estimated switch



The main takeaway here is methodological: the model can pick out *which* spontaneous trials had a clear switch and roughly *when*, purely from the loudness/accent pattern, without ever being told a switch happened.

A closer look: a small timing bump right after the switch

Timing wasn't reliable enough to detect a switch on its own (Part 1), but now that we have τ from the loudness model, it's worth looking at timing *around* that known point more closely than before. Using a narrow sliding window (much finer than earlier passes), there's a small, localised bump in timing error right after the switch: tap-to-click error dips exactly at the switch, then rises again for a few beats, before settling back down.



This is a small effect on one participant’s data, so it isn’t something to draw conclusions from yet – but it’s worth keeping an eye on once more participants are added: if it holds up, it would suggest a brief timing wobble a beat or two after the switch itself (distinct from the broader pre-vs-post trend), which could be a genuine signature of the participant re-locking onto the new meter right after adopting its accent.

Part 4: Comparing conditions and meters

Beyond `tau` itself, the model fits three other parameters per trial: `s` (how sharp the switch is), `sigma_sw` (extra noise right at the switch), and `sigma_0` (baseline loudness noise, unrelated to the switch). Do any of these differ between cued and spontaneous trials, or between meter 2 and meter 3? Comparing posterior *means* alone can be misleading – it ignores how uncertain each trial’s own estimate is. Instead, `02_fit_model.ipynb` computes a proper 94% credible interval on each group difference (by resampling posterior draws across trials within each group many times, building up a distribution of “the group’s mean” that respects each trial’s own uncertainty, then differencing). A difference only counts as credible if that interval excludes zero.

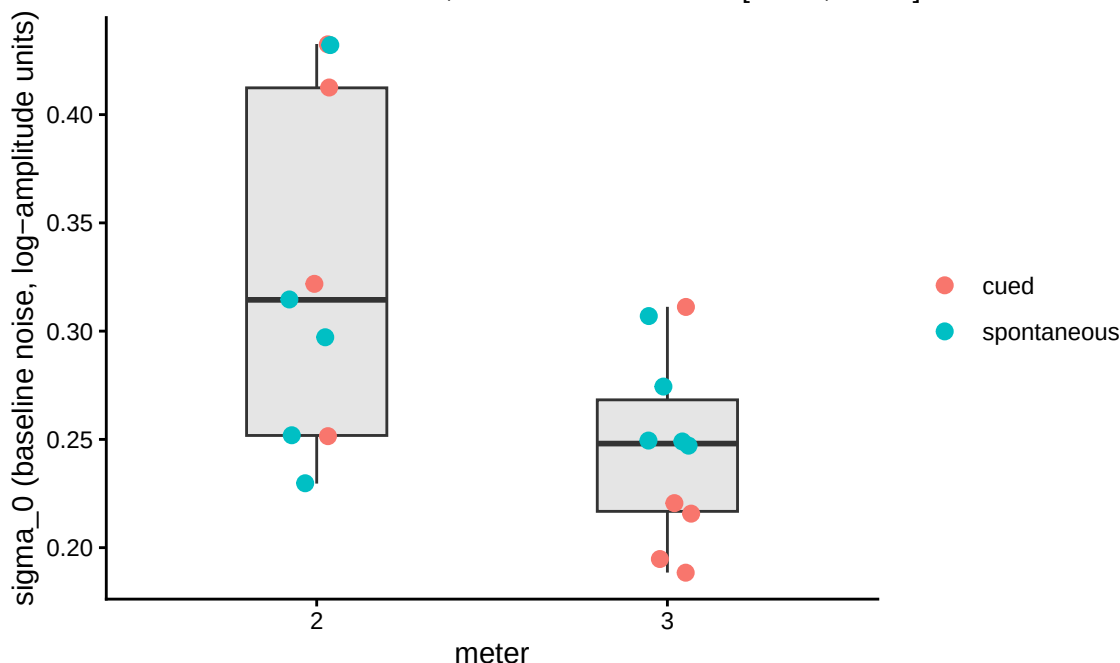
Comparison	Parameter	Group A mean	Group B mean	Difference	94% credible interval	Credible?
cued vs spontaneous	<code>tau</code>	27.546	22.827	4.719	[3.346, 6.005]	yes
cued vs spontaneous	<code>s</code>	0.167	0.164	0.003	[-0.058, 0.063]	no
cued vs spontaneous	<code>sigma_sw</code>	0.428	0.363	0.065	[-0.171, 0.321]	no
cued vs spontaneous	<code>sigma_0</code>	0.283	0.285	-0.002	[-0.029, 0.026]	no
meter 2 vs meter 3	<code>tau</code>	24.509	25.581	-1.072	[-2.425, 0.324]	no
meter 2 vs meter 3	<code>s</code>	0.168	0.162	0.006	[-0.055, 0.064]	no
meter 2 vs meter 3	<code>sigma_sw</code>	0.437	0.356	0.081	[-0.163, 0.318]	no
meter 2 vs meter 3	<code>sigma_0</code>	0.327	0.246	0.081	[0.052, 0.109]	yes

Two things stand out. `tau` differs credibly between cued and spontaneous – but that’s not a new result, it’s the same reaction-lag effect already reported in Part 2 (cued trials’ `tau` sits `switch_beat` + ~2.3 beats later; spontaneous trials have no `switch_beat` to add that to). `s` and `sigma_sw` show no credible difference anywhere: switches are equally sharp and equally noisy at the switch, regardless of condition or meter, as far as this data can tell.

The one genuinely new result is `sigma_0`: **meter-2 trials are noisier at baseline than meter-3 trials**, regardless of condition – consistent across nearly every individual trial, not driven by one outlier.

Baseline tapping noise is higher in meter-2 trials

meter 2 – meter 3 = 0.081, 94% credible interval [0.052, 0.109]



This is unrelated to the switch itself – `sigma_0` describes general tap-to-tap loudness consistency, not anything specific to *when* the meter changes. A plausible reading: a 2-beat bar gives the tapper less internal structure to lock onto (just strong-weak) than a 3-beat bar (strong-weak-weak), making steady, consistent accenting harder to sustain – but with one participant this is a hypothesis worth checking with more data, not a conclusion.

Part 5: How much data would we actually need?

Part 4’s comparisons came back mostly non-credible with the ~5 trials/cell we have now (meter 2/3 x cued/spontaneous). That’s ambiguous – “no real effect” or just “not enough data”? `power_analysis.ipynb` simulates new datasets at the same trial-to-trial noise seen in sub02’s fits to answer that; full method there, headline numbers below.

Parameter	Effect	Effect size	Standardized effect (Cohen’s d)	Trials/cell needed for 80% power
tau	meter	1x the observed diff	0.28	104
tau	meter	2x the observed diff	0.55	28
tau	meter	3x the observed diff	0.83	12

Parameter	Effect	Effect size	Standardized effect (Cohen's d)	Trials/cell needed for 80% power
tau	meter	5x the observed diff	1.39	6
tau	interaction	1x the observed diff	0.83	48
tau	interaction	2x the observed diff	1.67	12
tau	interaction	3x the observed diff	2.50	6
tau	interaction	5x the observed diff	4.17	4
s	meter	1x the observed diff	0.65	20
s	meter	2x the observed diff	1.30	6
s	meter	3x the observed diff	1.94	4
s	meter	5x the observed diff	3.24	4
s	condition	1x the observed diff	0.40	50
s	condition	2x the observed diff	0.79	14
s	condition	3x the observed diff	1.19	8
s	condition	5x the observed diff	1.98	4
s	interaction	1x the observed diff	1.14	26
s	interaction	2x the observed diff	2.28	8
s	interaction	3x the observed diff	3.42	4
s	interaction	5x the observed diff	5.70	4
sigma_0	meter	1x the observed diff	1.27	6
sigma_0	meter	2x the observed diff	2.54	4
sigma_0	meter	3x the observed diff	3.80	4
sigma_0	meter	5x the observed diff	6.34	4
sigma_0	condition	1x the observed diff	0.03	NA
sigma_0	condition	2x the observed diff	0.06	NA
sigma_0	condition	3x the observed diff	0.09	NA
sigma_0	condition	5x the observed diff	0.15	NA

Parameter	Effect	Effect size	Standardized effect (Cohen's d)	Trials/cell needed for 80% power
sigma_0	interaction	1x the observed diff	1.39	18
sigma_0	interaction	2x the observed diff	2.78	6
sigma_0	interaction	3x the observed diff	4.17	4
sigma_0	interaction	5x the observed diff	6.94	4

Most effects *actually observed* so far need well more than the current ~5 trials/cell to reach 80% power (e.g. 104 for **tau**'s meter effect), dropping fast for a bigger true effect (12 at 3x) – so most Part 4 non-findings are inconclusive, not evidence of no effect. **sigma_0**'s meter effect needed no such adjustment (already ~80% powered at n=5), which is a useful sanity check on the method. One new result: the **meter x condition interaction on tau** (d = 0.83) is bigger than either main effect and was never tested before – worth a dedicated follow-up (~48 trials/cell to confirm).

Suggestion: ~20 trials/cell (80/participant) as a practical default, enough for a moderate-to-large effect (2-3x observed) on most parameters above; size specifically to the **tau** interaction row if that's the actual target.

Separately: if switch “selectivity” ($-\log(s)$) scales with musical training, how many *participants* would that take? (Needs an assumption about between-participant variance, which one participant can't measure – see the notebook.)

True correlation (r)	Trials/participant	Participants needed for 80% power
0.3	10	92
0.3	20	88
0.3	30	88
0.5	10	32
0.5	20	30
0.5	30	30
0.7	10	16
0.7	20	14
0.7	30	14

Participant count is the real lever here, not trials/participant (which barely moves the requirement): a moderate effect (r = 0.5) needs roughly **30 participants**, a strong one (r = 0.7) ~**15** – keep sessions around the current length and prioritise recruiting more participants.

Summary

- Tapping timing looks noisy and undifferentiated around a switch, both in a single trial and pooled across all of them – **loudness, not timing, is where the switch signal actually lives**, which is why the model is built entirely on tap loudness.
- The switch-detection model was validated against cued trials (known instruction time) before being trusted on spontaneous trials: **tau** tracks the instruction time closely (correlation 0.94) with a small, consistent, always-positive lag (~2.3 beats on average) – a sensible reaction/adoption delay, not the cue time itself, which is what the model is actually built to measure. Real switches also turned out to be close to instantaneous rather than gradual, once checked directly.

- For spontaneous trials, the model both estimates *when* a switch happened and *how confident* it is in that estimate – 10 in 10 spontaneous trials show a confident, well-identified switch.
- A model-free check (raw accent contrast, no fitting) confirms the pattern independently: confident trials show a real contrast collapse near the model’s estimated switch; unclear trials show either a weak starting accent or an incomplete collapse – genuine data features, not fitting artifacts.
- All results so far are based on a single participant; more data is needed before drawing general conclusions.
- A power analysis of the full 2x2 (meter x condition) design suggests roughly **20 trials/cell (80/participant)** to properly resolve most **tau/s** effects at 2-3x their currently-observed size, and hints at a **tau meter x condition interaction** worth a dedicated follow-up. Across participants, roughly **30 participants** (session length largely unchanged) would be needed to detect a moderate musical-training effect on switch selectivity.